

Role Support in Knowledge-Graph Error Ranking: Predictor Regimes and Evaluation Policies

Active revision | Summary: 27 September 2026

When does knowledge-graph role support help rank prediction errors, and how stable is its meaning across predictor regimes?

CENTRAL IDEA

Evaluate support counts alongside confidence, frequency and ensemble disagreement. Keep the trained predictor and the candidate-evaluation policy explicit instead of treating missing role support as an intrinsic uncertainty score.

MAIN FINDING

As observed-target accuracy improves from 10.05% to 44.35% in the studied ComplEx comparison, the mean error AUROC of the raw missing-role score reverses from 0.689 to 0.364. Continuous counts improve AURC and Brier score in all 12 reported ensemble/error-head comparisons.

EVIDENCE AND SCOPE

The revision examines 12 from-scratch DistMult/ComplEx predictors, six development-selected continuation stages and six fixed three-member ensembles on two static graphs. Candidate policy and a fixed coverage threshold can change apparent headroom and rankings.

LIMITS

Binary missing-support bits are deterministic functions of counts; any fitted gain is not new information. Results concern recorded graph targets, not calibrated factual truth. Development reuse, shared graph structure and limited seeds restrict confirmation and causal interpretation.

TAKEAWAY

Role support is useful context for error diagnosis, but its direction and practical value depend on the predictor and evaluation policy.

한국어 핵심 요약

지식 그래프에서 역할별 관측 횟수는 오류를 진단하는 데 도움이 될 수 있습니다. 그러나 관측이 없다는 사실을 언제나 높은 오류 위험으로 해석할 수는 없습니다. 학습된 예측기와 후보 평가 정책이 달라지면 방향과 성능도 바뀝니다. 기록된 정답의 오류 순위와 사실 자체의 확실성을 구분합니다.

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